

Fine-mapped eQTL evaluation using XQTL mega-eQTL data

1 Overview

This analysis applies the variant-effect benchmark in `eqtl.tex` to six cell-type mega-eQTL fine-mapping datasets supplied as Sum of Single Effects (SuSiE) posterior inclusion probabilities (PIPs) for expression quantitative trait loci (eQTLs). In this report, XQTL names this six-group source. The groups are astrocytes (Ast), excitatory neurons (Exc), inhibitory neurons (Inh), microglia (Mic), oligodendrocyte precursor cells (OPC), and oligodendrocytes (Oli). The input does not contain an endothelial group. We compare these results with the existing singlebrain fine-mapping, using its broad Ast, Ext, IN, MG, OPC, and OD panels, respectively.

2 Methods

For both data sources, gene identifiers are version-normalized and matched to the same GENCODE hg38 gene annotation. Variants are restricted to single-base ACGT substitutions on the gene chromosome within 512 kb of the gene transcription start site (TSS). Variants with $\text{PIP} > 0.75$ are positives; those with $\text{PIP} < 0.01$ are negatives. The benchmark samples up to 1,000 variants from each class in each group with a fixed seed. Each model receives a 1 Mb sequence window centered at the gene TSS.

The trained comparisons reuse the head-only, LoRA, and LoRA+LoCon checkpoints and RNA track mappings in `eqtl.tex`. Native AlphaGenome mutation effects use all available RNA-seq and CAGE tracks, followed by the same biologically constrained track selection used in that report. Performance includes area under the receiver operating characteristic curve (AUROC) and average precision (AP), overall and by distance to the TSS; random AP equals the positive prevalence within each distance bin. CAGE means Cap Analysis of Gene Expression.

Fine-mapping agreement is computed on gene-variant keys consisting of gene, chromosome, one-based position, reference allele, and alternate allele. Exact high-confidence overlap uses all eligible variants and reports intersection, Jaccard similarity, and directional recall at $\text{PIP} > 0.75$. Continuous PIP agreement is assessed on a seeded sample of up to 25,000 eligible rows per group, matched by the same gene-variant key; Pearson and Spearman correlations are reported for shared sampled rows. The PIP sample is used for correlation and visualization only, not for the exact high-PIP overlap.

3 Fine-mapping agreement

The full scan covered 70.9–218.7 million input rows per group. After matching to an annotated gene, requiring a single-base substitution on the gene chromosome, and applying the TSS window, 14.3–41.2 million records per group remained. Table 1 gives counts and PIP classes.

At the high-confidence threshold, exact gene-variant Jaccard similarity is 0.080–0.118 (Table 2). XQTL positive recall against singlebrain is 0.28–0.37, while singlebrain positive recall against

Table 1: XQTL source counts. Eligible rows satisfy the common gene, allele, chromosome, and TSS-window filters.

Group	Input rows	Eligible rows	PIP > 0.75	PIP < 0.01
Astrocyte	167,399,779	31,850,543	581	31,769,295
Excitatory	218,708,555	41,150,681	1,784	40,976,915
Inhibitory	207,868,448	39,394,880	813	39,281,800
Microglia	70,865,864	14,332,701	168	14,305,646
OPC	122,683,778	24,317,596	358	24,269,005
Oligodendrocyte	178,552,659	33,830,282	852	33,738,029

XQTL is 0.09–0.16. Thus the sources share some high-confidence calls but the two sets are not interchangeable. On up to 25,000 randomly sampled eligible XQTL records per group, 9,238–12,359 have an exact gene-variant match in singlebrain. Their Spearman PIP correlations are 0.16–0.20, indicating weak rank agreement. Pearson correlations are higher in some groups because of the strongly zero-inflated PIP distribution and should not be interpreted as uniform agreement across the PIP range.

Table 2: Exact high-PIP overlap and sampled continuous-PIP agreement. Recall columns use the named source as denominator.

Group	XQTL n	Singlebrain n	Shared	Jaccard	XQTL recall	SB recall	Spearman r
Astrocyte	581	1,231	172	0.105	0.296	0.140	0.177
Excitatory	1,784	3,420	504	0.107	0.283	0.147	0.194
Inhibitory	813	1,838	236	0.098	0.290	0.128	0.187
Microglia	168	668	62	0.080	0.369	0.093	0.196
OPC	358	929	122	0.105	0.341	0.131	0.158
Oligodendrocyte	852	1,820	283	0.118	0.332	0.155	0.172

4 Variant-effect performance

The model and native-track runs are pending; performance results will be inserted when scoring is complete.

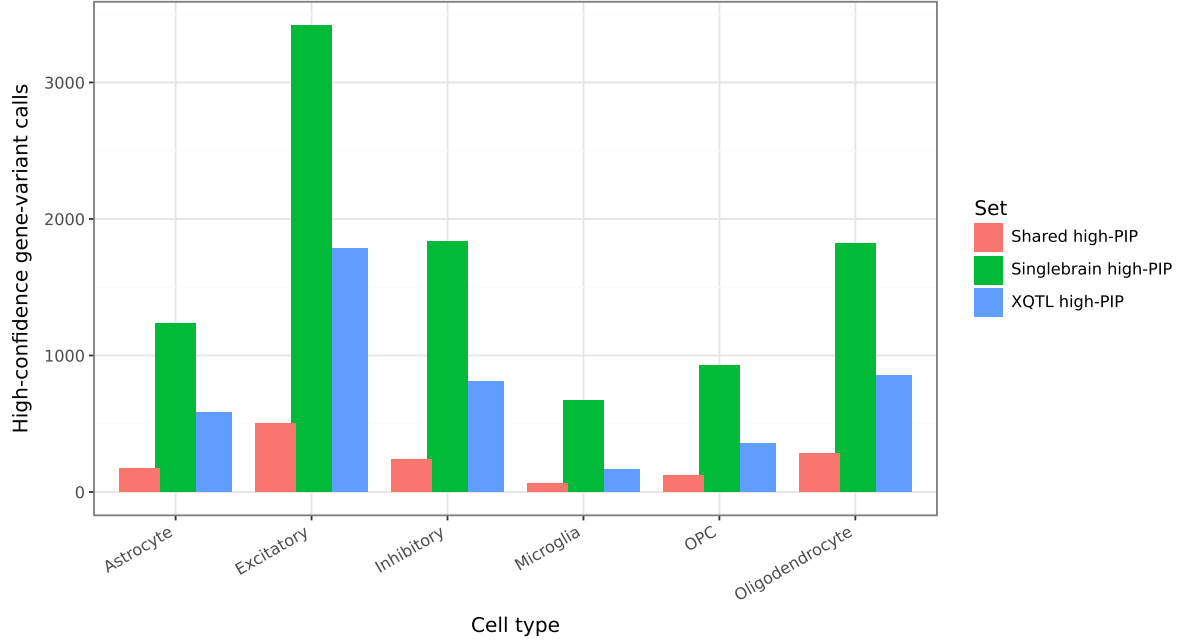


Figure 1: Counts of eligible high-PIP gene-variant calls in each source and their exact intersection.

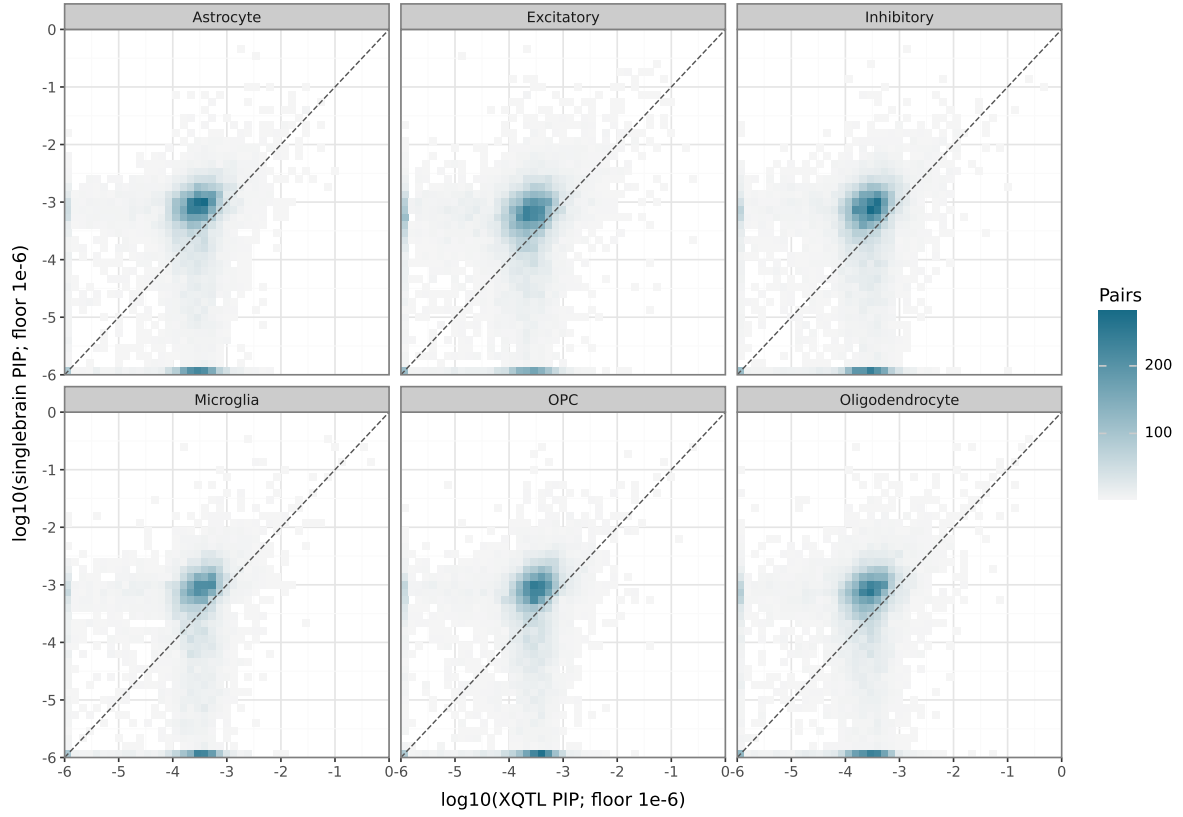


Figure 2: Continuous PIP agreement for sampled gene-variant pairs present in both sources. Axes show $\log_{10}(\max(\text{PIP}, 10^{-6}))$; the dashed diagonal indicates equality.